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Roll No

MI-7004(2)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Surface Mining Planning

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe the various methods and layout of waste dump construction in Coal mines.
b) Explain the horizontal tandem operation by Dragline with suitable layout. Under which condition, this method is preferred.
2. a) Explain vertical tandem operation of Dragline by drawing suitable layout.
b) Describe a Bucket wheel excavator with a neat sketch under the following headings:
 - i) Constructional features
 - ii) Wheel design
3. a) Discuss the role of following in open pit mine production planning:
 - i) Breakeven cut off grade
 - ii) Ultimate pit slope

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- b) Determine the value per tonne of copper ore and the breakeven stripping ratio for the following data:
Grade-3% Cu,
Overall recovery-90%,
Total cost of production of mineral-Rs5000/Tonne of ore
Selling Price-Rs 300 /kg of Cu
Cost of stripping- Rs 70/m³
4. a) Write down the objectives of Production planning and significance of cut off grade.
b) What are the various factors affecting the design of Ultimate Pit? Explain manual method of designing ultimate pit.
5. a) What is meant by stability of slopes and slope failure? Discuss the importance of pit slope angle on the economics of open pit mining operation.
b) Discuss the various factors influencing the stability of waste dump slopes.
6. a) Explain the techniques used for analysis of slopes under circular failure condition.
b) Discuss the various geotechnical factors influencing the stability of hard rock slopes.
7. a) Explain how you will ensure proper drainage of water around slopes for their stability enhancement. Draw suitable diagram of drainage arrangement.
b) Discuss the various Geometrical conditions under which circular and wedge failure of slopes may occur.
8. Explain the techniques used for analysis of slopes under circular failure condition.

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